

1. FizzBuzz Challenge Write a program that iterates from 1 to 50. - Print 'Fizz' for multiples of 3 - Print 'Buzz' for multiples of 5 - Print 'FizzBuzz' for multiples of both
2. Leap Year Logic Write a function that takes a year as input and returns True if it is a leap year and False otherwise.
3. Identity vs Equality Given: L1 = [1,2,3] and L2 = [1,2,3] Explain the difference between L1 == L2 and L1 is L2.
4. Bitwise Swap (Advanced) Swap two numbers a and b without using a third variable or multiple assignment. Use XOR operators. Section 2: Control Flow & Loops
5. Prime Number Finder Write a while loop that finds and prints the first 10 prime numbers.
6. The Loop Breaker Create a loop from 1 to 20: - If divisible by 4 → pass - If the number is 13 → continue - If the number is 18 → break
7. Nested Conditionals Write a program that takes a numerical grade (0–100) and returns: A (90+), B (80–89), C (70–79), Fail (below 70).
8. Reverse an Integer Using a while loop and arithmetic operators (// and %), reverse the digits of an integer. Section 3: Data Structures (Lists, Tuples, Sets)
9. Runner-Up Score Given a list of scores, find the second-highest score without using sort() or sorted().
10. Duplicate Removal Write a single line of code to remove duplicates from a list and return unique values.
11. List Intersection Write a program to find common elements between two lists using the membership operator in.
12. Tuple Immutability Test Attempt to change the second element of a tuple and handle the resulting error using try-except. Section 4: Dictionaries & Comprehensions
13. Character Frequency Counter Take a string input and return a dictionary with character counts.
14. List Comprehension Create a list of squares of all even numbers between 1 and 20 using a single line.
15. Dictionary Comprehension From the dictionary {'a': 1, 'b': 2, 'c': 3, 'd': 4}, create a new dictionary containing only items with values greater than 2